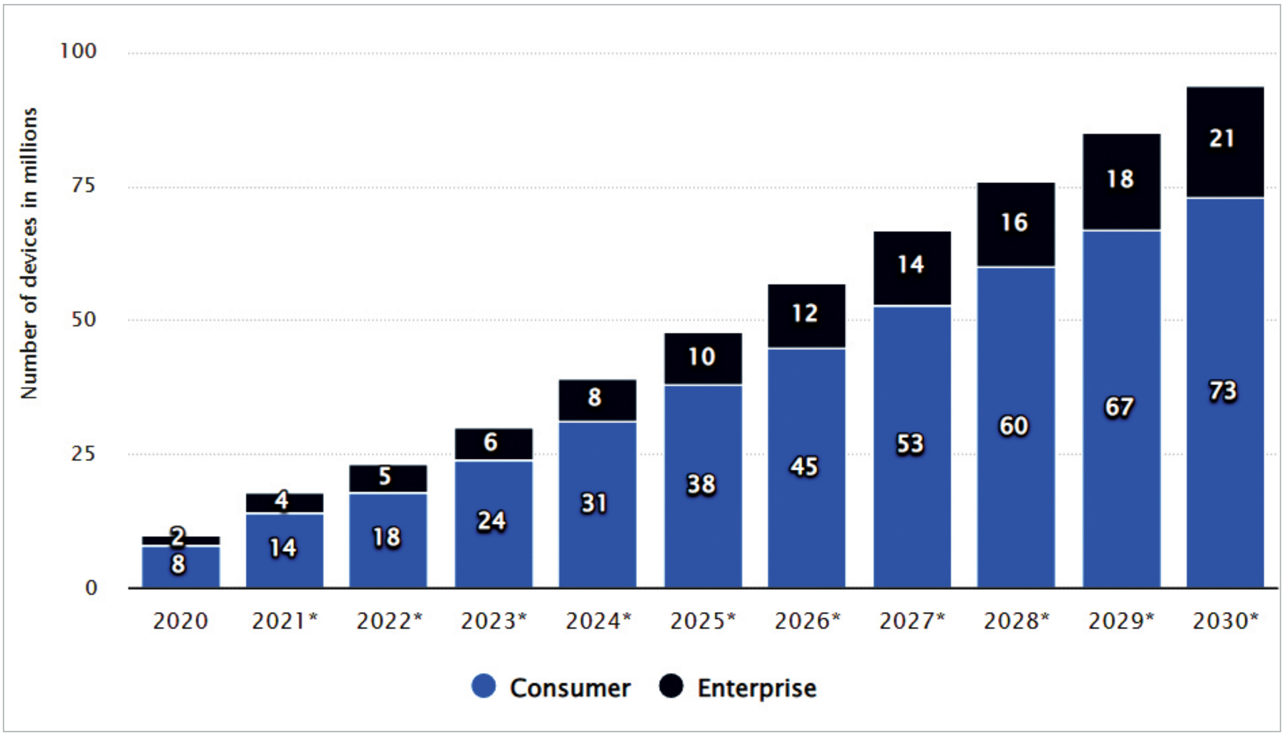

Figure 1: Market analytics for AR and VR devices (2020-2030) (*https://www.statista.com/statistics/1259882/human-machine-interface-augmented-virtual-reality-devices-worldwide*)

Architects, builders and real estate companies can utilise VR to generate a virtual tour of a new building or layout, while AR can help to illustrate the structure and systems of a building superimposed over a real-world scenario.

Key implementation domains in AR and VR

AR and VR have a wide and huge scope in education, e-governance, the corporate sector and many other domains, as listed below:

- Architecture or building plans
- Qualitative analysis in chemistry
- Assembly/disassembly of an electrical engine
- Exploration of human anatomy
- Simulation of circulation of blood in heart
- Simulation of flight/vehicle/space stations
- Building electronic circuits in zero physical defects
- Virtual classrooms for teaching and learning

Myths regarding AR and VR implementations

AR and VR can be implemented using minimum resources as well as very limited infrastructure and hardware. There are a lot of myths associated with AR and VR, some of which are listed below:

- Need for huge resources and costly hardware
- Specialised graphic cards are needed
- Specialised processors are needed
- Integration of GPU/TPU is required
- Need for high performance programming frameworks

None of the above is true.

Key features required for effective use of AR and VR

To reap the maximum benefits of AR and VR, a few key features must be integrated into the devices, including software and hardware infrastructure:

- Drag-and-drop
- Dynamic editing
- Simulation
- Real-world backdrop
- Superimposed objects
- Hardware integration
- SLAM (simultaneous localisation and mapping) support
- Cross-device publishing
- Real-time analytics
- Uploading dynamic content
- 3D object tracking
- Support for smart glasses
- Content creation
- Geolocation tracking
- Cloud storage

A few open source high performance platforms used for implementing AR and VR are listed below.

A-Frame *(aframe.io):*

- Open source platform for VR with entity component system
- Java based programming functionality for high performance implementations
- No need for any third party application/hardware
- All we need is a Web browser and JavaScript library